



Stan and Davis Bacchus Winery

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&

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01. Overview

Stan and Davis Bacchus inherited their winery from their father, George, three years ago upon his retirement. They are seeking outside advisement counseling to incorporate new business methods to help them improve their products and customer service.

Stan and Davis know the yearly 'snapshot' of the business is due in a few weeks, and need the information to determine the state of the inventory, the distribution and the employees. They need information on: 1. Are all suppliers delivering on time? 2. Which distributor carries which wine? 3. Employee time. During the last four quarters, how many hours did each employee work?

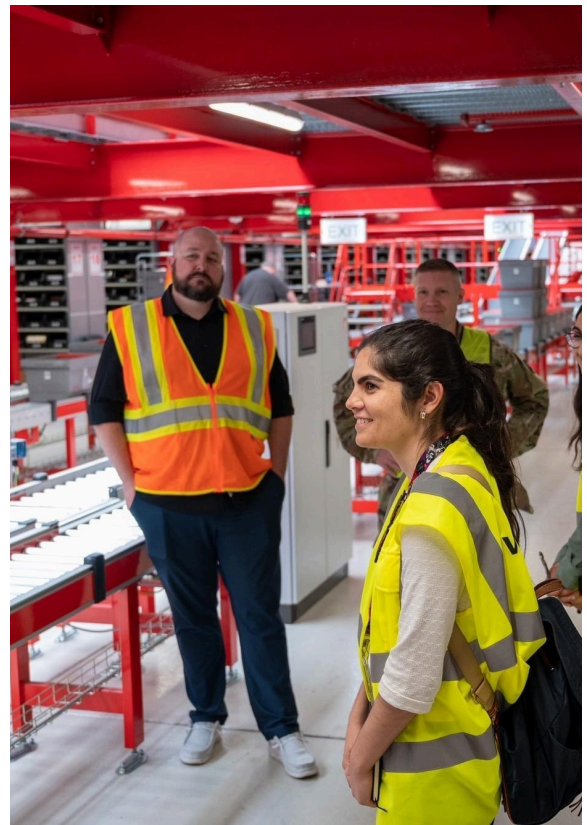


02. Key findings

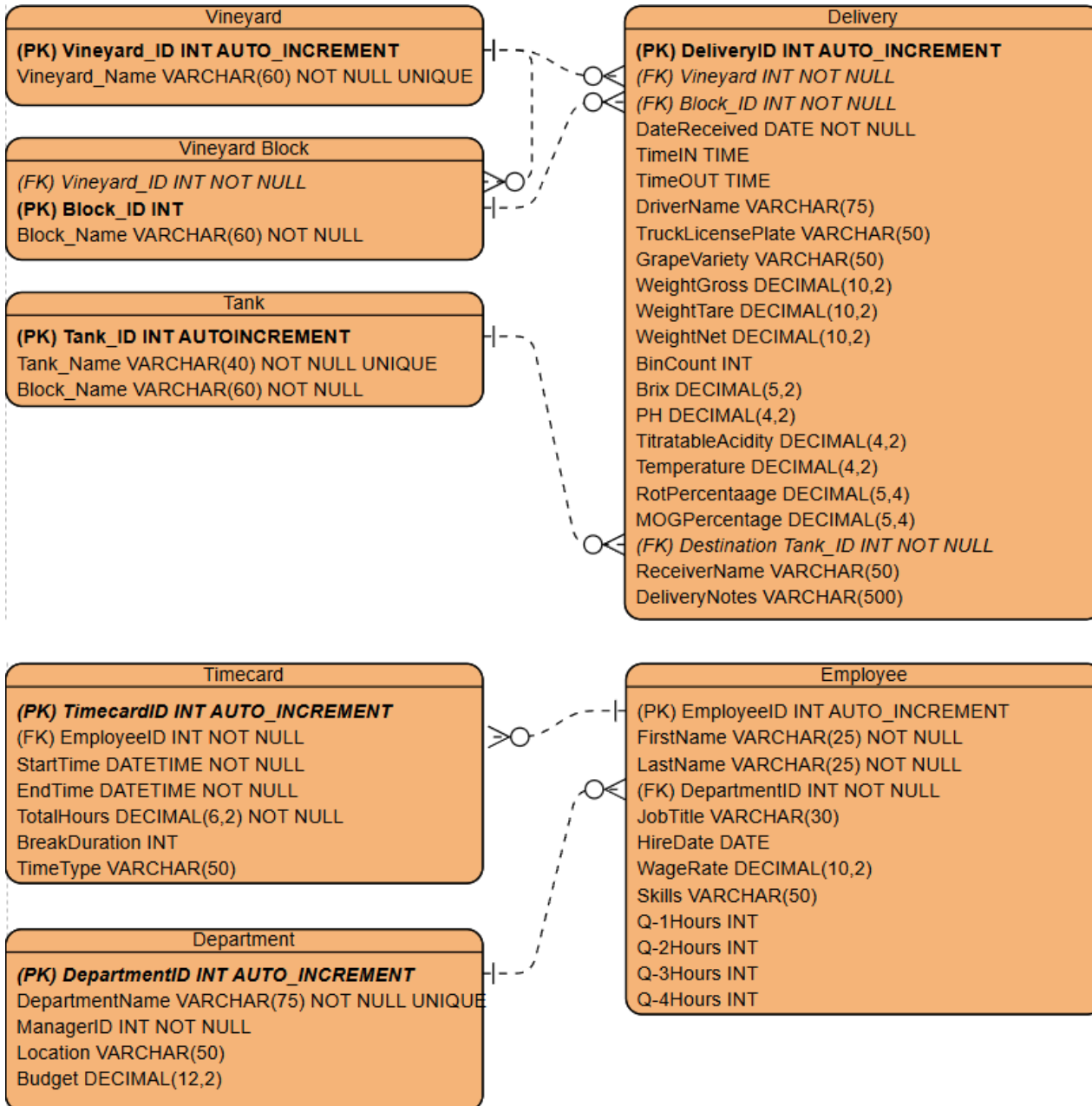
- a. There is a delivery tardy length beyond 5 days at a rate of 33% from the most recent snapshot of collected deliveries.
- b. Current distributor bins have a stock greater than 250 of each wine they are carrying but there is a clear lack of variety for each bin.
- c. Median employee hours put in over the last year are 1,950. This averages 16.25 days PTO use per year per employee.

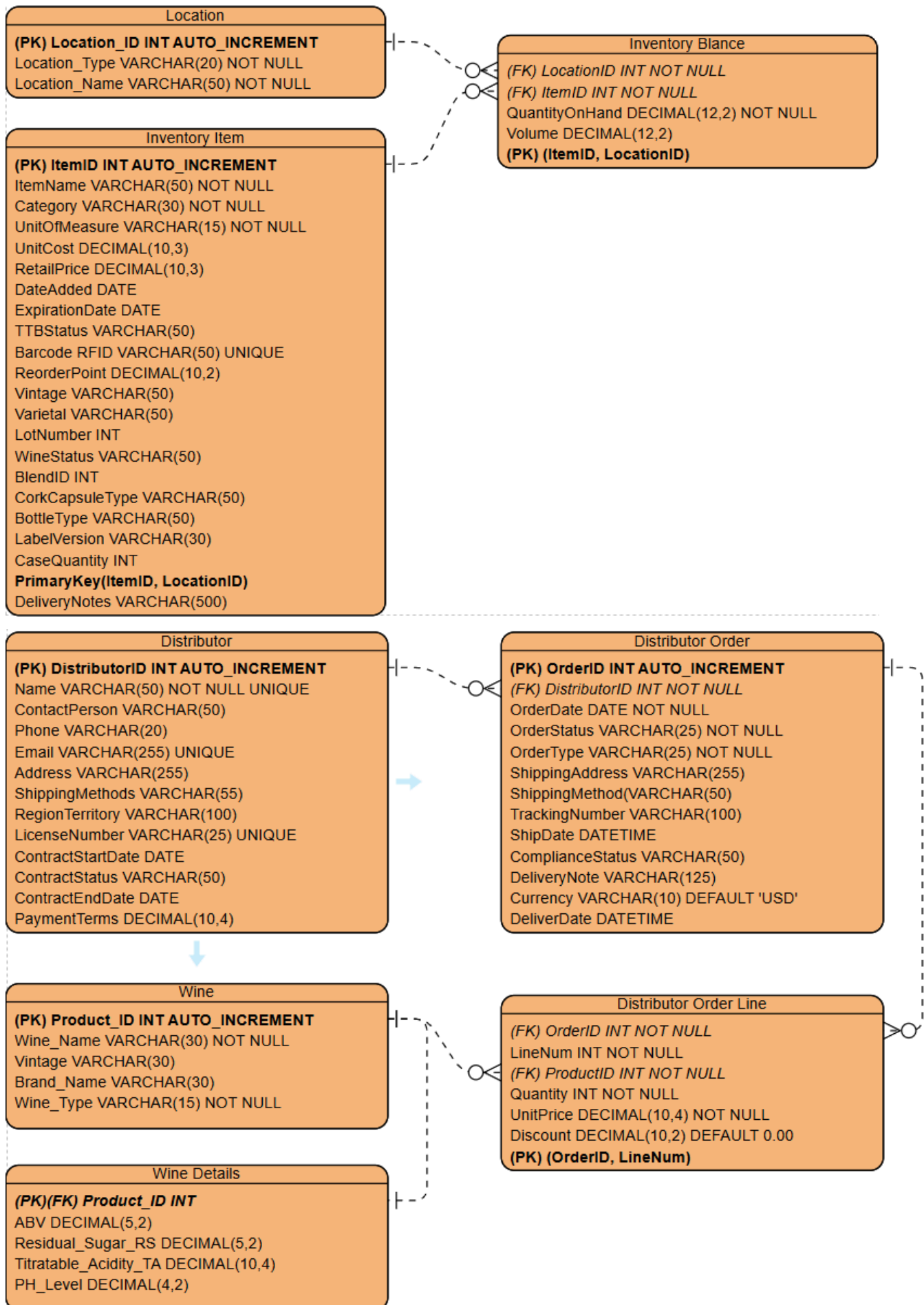
03. Solutions

- a. Review supply and distribution contracts for delivery requirements. 33% tardy rate falls far outside of standard distribution rates and may be cause for business risk.
- b. Database solution for tracking business operations created. This modernizes without rewriting business practices.
- c. Employee PTO usage falls within standard use and may even be low. Employee retention is high value and this may be a point for concern.



Final ERD





Analysis



```
-- DISPLAYING DELIVERY STATUS --  
Tracking Number: TRK-904  
Delivery Time in Days: 7  
Status: LATE  
  
Tracking Number: LTL-9468  
Delivery Time in Days: 7  
Status: LATE  
  
Tracking Number: TRK-701  
Delivery Time in Days: 2  
Status: OK  
  
Tracking Number: TRK-902  
Delivery Time in Days: 2  
Status: OK  
  
Tracking Number: CR-120  
Delivery Time in Days: 4  
Status: OK  
  
Tracking Number: TRK-907  
Delivery Time in Days: 3  
Status: OK
```

01. Delivery Timeliness

This report goes over completed deliveries and gives a status based on the number of days it took for delivery ordered by status giving 'LATE' the priority in attention. 'LATE' is determined to be any delivery over 5 days as that is standard business delivery standards (Despite what Amazon has conditioned you to).

02. Wine Distribution

The second report goes over wine inventory by location. Further improvements to this report would be to iterate through location groupings and pair wines to locations so that you could more easily see BinA1 carries both Merlot and Chardonnay. With 4 wine varieties available and only 2 locations using 1 or 2 of them shows a clear lack of distribution variety.

```
-- DISPLAYING WINE LOCATIONS --
Locale: Bin A1
Wine: Merlot 2023 (Finished)
Amount: 320.00 bottle(s)

Locale: Bin B2
Wine: Cabernet 2023 (Finished)
Amount: 280.00 bottle(s)

Locale: Bin A1
Wine: Chardonnay 2023 (Finished)
Amount: 260.00 bottle(s)
```

```
-- DISPLAYING EMPLOYEE HOURS BY QUARTER AND TOTAL --
Emp ID: 10001
Q4: 480
Q3: 500
Q2: 510
Q1: 500
Total: 1990

Emp ID: 10002
Q4: 475
Q3: 510
Q2: 505
Q1: 490
Total: 1980

Emp ID: 10003
Q4: 480
Q3: 510
Q2: 510
Q1: 440
Total: 1940

Emp ID: 10004
Q4: 460
Q3: 490
Q2: 480
Q1: 500
Total: 1930

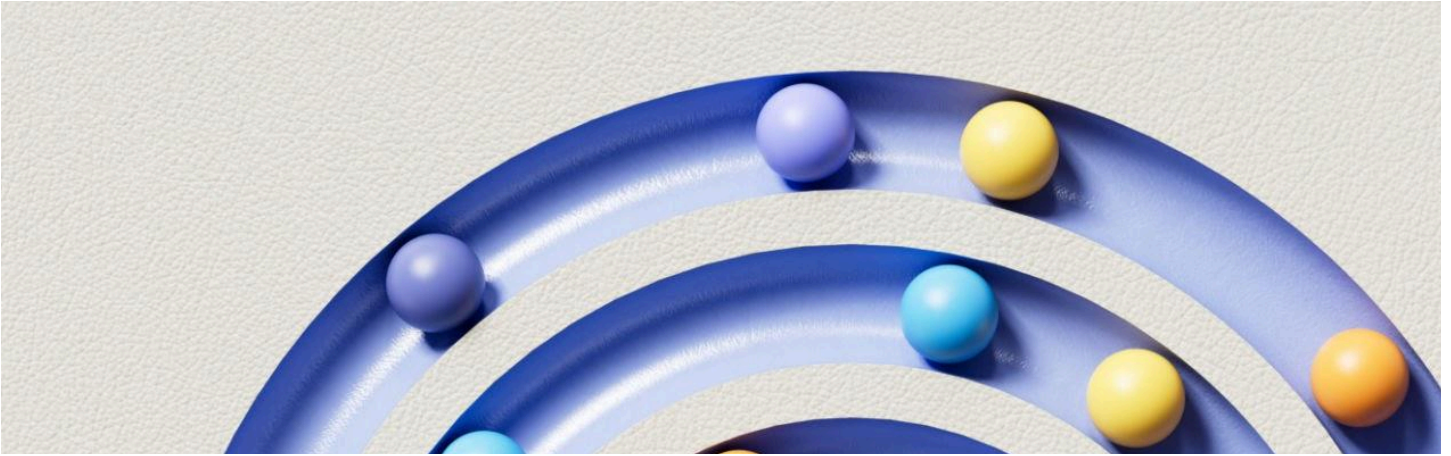
Emp ID: 10005
Q4: 470
Q3: 510
Q2: 425
Q1: 510
Total: 1915

Emp ID: 10006
Q4: 480
Q3: 500
Q2: 120
Q1: None
Total: 1100
```

03. Employee Hours

The last report goes through each employee and gives their hours worked by quarter iterating back then totalling the last 4 quarters. A standard full time schedule is 2080 hours. Omitting employee 10006 as they started in Q2 and haven't served a full year this averages ~1,950 hours worked for a full year per employee. That amounts to 16 days PTO used per year. The industry standard is closer to 20 which raises the question of incentivising use of PTO as burnout and turnover costs are often greater than any PTO used by trained and seasoned employees.

Assumptions



“Items” in inventory (bottles, Corks) are tracked by type and quantity, not individually by SKU.

Rather than holding onto a year’s worth of timestamps, the quarters are set up as follows. The “quarterly snapshot” assigns the recent quarter’s timecards to Q1. From there each new quarter is assigned back to front where Q4 gets assigned Q3 counts, then Q3 gets Q2 and Q2 gets Q1. Finally Q1 will get the total of timecard hours assigned to each employee.

“Wine distribution” assumes the winery does not sell directly to consumers, only to distributors.

Time tracked for employees are by the initial 6 employees mentioned in the Case Study. The 20 employees managed by Henry Doyle are general production staff, and their hours are tracked individually.

Only 3 of the 4 types of wines are being used by distributors as a Findings example. Another Findings example included is a high rate of LATE deliveries quantified by greater than 5 days delivery time.

Distributor assumption is the count values match what Bin values we used.